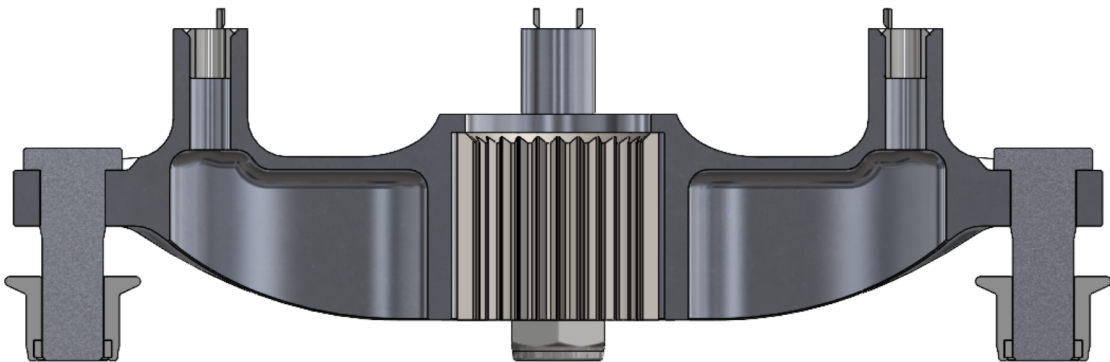




Cornell University Baja Racing Team

Spring 2026

Front & Rear Hubs



Technical Report

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1 Hubs

1.1 Abstract

The the front and rear hubs allow the driveline to apply torque to the ground and they react the loads seen at the tire to the rest of the suspension set up. Additionally the front hubs mounts the break rotors and must react break locking loads. The primary design challenge for the hubs is reacting complex combined loading while optimizing for weight savings.

1.2 Design Requirements

Constraints:

- Must be able to transfer max driveline torque
- Must not break under any typical suspension load case
- Must not fail due to fatigue for competition cycle
- Must integrate with OEM hardware

Objectives:

- Redesign torque transfer method
- Remove OEM parts
- Maintain manufacturability

1.3 Initial Research

1.3.1 Last Year's Car

TG21 is the first car to use the front banana hubs design which pushes the wheel hub mounting plane inboard to reduce KPO (king pin offset). On TG21 the KPO was 2" requiring a significant banana shape. On TG21 kept with the legacy design of using OEM Yamaha hubs that had their arms cut off and machined into an insert.

1.3.2 OD16

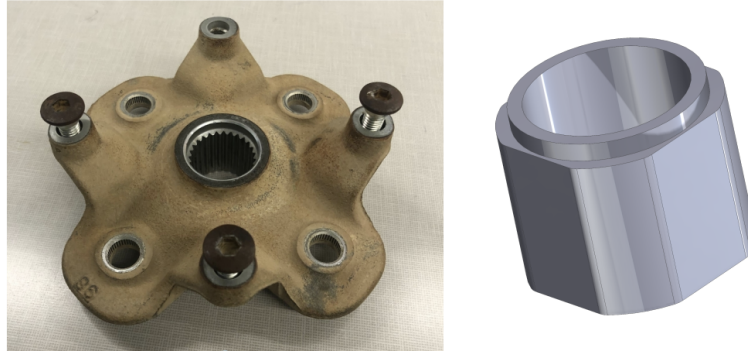


Figure 1: Yamaha Hub and OD16 insert

Since OD16 we have used a spine insert for the hubs that is harvested from a OEM Yamaha hub. That spline insert design has not been significantly updated since OD16.

1.3.3 Online Research

Other baja teams use a wide range of different designs for their hubs. There is a lot of functionality differences such as having outboard sprags, and teams that use aluminum splines for the hubs.

1.4 High Level Description

The primary change for TG22's hubs is the removal of the OEM insert design. TG22 had a 15% increase in torque to both its front and rear driveline components requiring the hubs and insert to withstand higher torques. To accomplish this I reversed engineered the OEM spline by referencing JIS standards, and tweaking the profile using 3D resin printed test parts. The insert torque transfer profile had not been updated since OD16 and consisted of a square with rounded corners. For TG22 the torque transfer profile was updated to be a p4c polygon initially designed using hand-calcs then further analyzed in Ansys.

1.4.1 Hub Spline Inserts

The spline insert has two main components to it. The outer p4c torque transfer profile and the inner spline profile. I used ISO spline standards to calculate the FOS for the reversed engineered spline profile coming up with a FOS of 1.46. The ISO spline standard is really conservative with KISSsoft having a 1.7 FOS using the Niemann/Winter Method which is only well documented in German. It seems valid but I can't read German and my attempts to understand it only partially succeeded.

The p4c polygon profile was designed using the teams torque transfer polygon spreadsheet developed for TG21's tri-lobes. Originally, the design was significantly more rounded

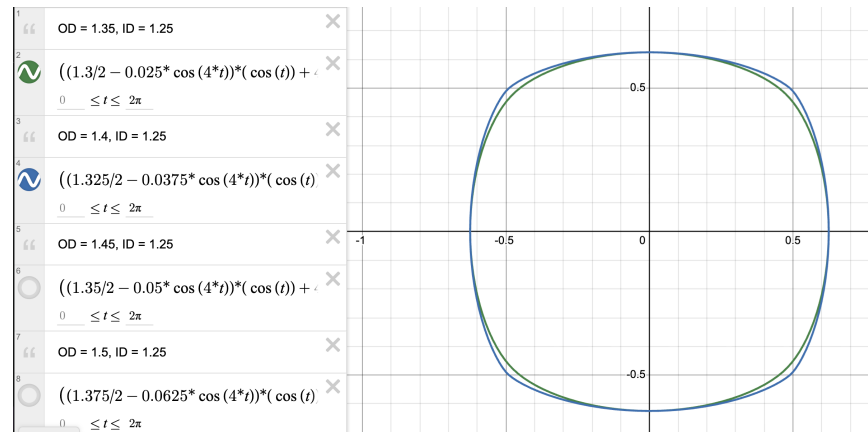


Figure 2: Desmos profile plot

to reduce weight and, according to the calculator, had an FOS of 1.6. After discussion, however, I chose to increase the eccentricity to account for manufacturing variations and better performance under combined loading, which resulted in an FOS of 2.34. I plotted the parametric profile in [Desmos](#) to better understand what the profile looked and make sure it passed a vibe check. Once I had the general size and shape of the profile I made sure it would be manufacturable by ensuring that the minimum radius of curvature was no less than 0.13 inches. This allowed for the speed tooling to manufacture the hub bore without any issues.

To integrate the insert into the Hub I used Ansys as well as relaying on legacy design solutions. The hand calculations specify a required minimum hub thickness for the polygon profile, but that would significantly increase the mass of the hub. To solve this problem I analyzed the legacy insert press in Ansys under different loading conditions, and compared the results to the new profile. The old profile had significant stress risers that had not caused issues for the last seven years, so I made sure that the stresses in the TG22 hub were equal or less than the legacy design. To maintain confidence I also kept the minimum wall thickness around the insert the same as the legacy design.

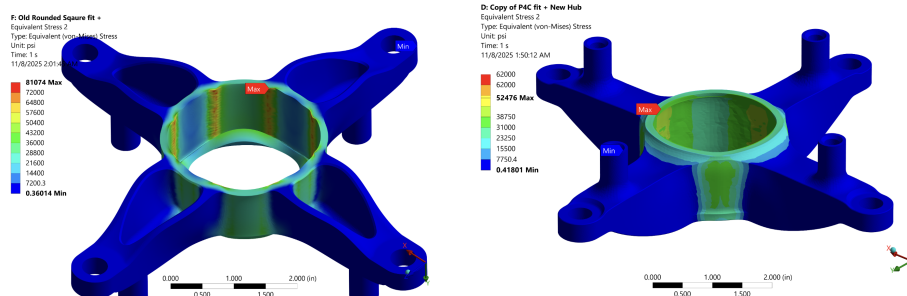


Figure 3: Ansys of TG21(FOS 0.88) & TG22(FOS 1.37) Press.

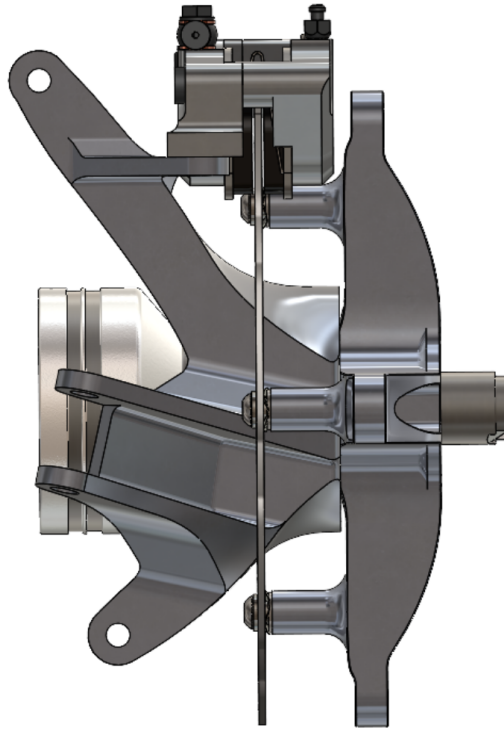


Figure 4: Front Wheel Assembly

1.4.2 Front Hub

The front hubs retained their banana shape to account for the KPO and even got more bananas because the KPO decreased from 2.1 to 2 inches. This means pushing the hub-rim interface closer inboard. I changed the limiting load case to landing with full driveline torque in a curb slam. Changing the limiting load case, increasing in driveline torque, and the new offset required me to re-optimize the hub arms to pass FEA. The corner had a lot of package issues because of the reduced KPO the wheel well got even tighter. As you can see in the front wheel assembly it is very tight clearances. To allow for proper clearances the OEM studs that are pressed into the hubs had to be turned down by $\frac{1}{50}$ thou.

1.4.3 Rear Hub

The rear hubs did not change too much even when taking into account the increase torque and high load case standard. FEA in Ansys was done to insure that the design was optimized for weight under our given load case

1.4.4 Manufacturing considerations

To accomplish my overall object of maintaining manufacturability the front and rear hub bodies where designed to be manufactured in house on 3-axis Hass machines. The press tolerance between the front and rear hubs where significantly different. The front hubs where a lot easier to press which suggests that they where made closer to nominal rather than with the tighter press.

The new insert was designed to so that only the wire EDM operation needed to be outsourced. When checking the hub assembly with its mating suspension components I noticed that there was a 60 thou gap between the hub's face and the bearing. To fix this the hub insert assembly was posted OPed in the Hass to deepen the splines by approx 70 thou. Only deepening the spline did not work, and slight radial clearances had to be made around the spline. There was another issue that caused a lot of headache was one



Figure 5: Post machined insert

insert's splines was not cylindrical with the outer profile of the part. This was noticed when post-oping the inserts in the Hass because the new machining was not concentric with the spline profile. At first we thought that the insert wasn't fully pressed in and slightly cocked eye in the hub body. Tragically this resulted in me breaking the hub when using the hydrologic press. When attempting to press the insert in more I did not place a support block underneath hub, so it was only supported by the rotor towers. Only after that indecent did I check the insert on a fixed cup which clearly showed that the hub was angled. Due to very limited timelines before competition that hub was labeled "CHUB" and would be used as a last resort spare.



Figure 6: Broken Hub



Figure 7: Chud Hub

1.5 Finite Element Analysis

1.5.1 Load Cases

I significantly updated the hub load cases this year to take into account the combined loading of the hub transferring torque while undergoing a curb slam event. I also included the p4c press mechanics in the analysis by doing a non linear multi step solve of the press then additional hub loading. While I added these new load cases I still relied on past load cases as points of comparison since we have not had a hub failure for years.

- Press: Non linear analysis of the press mechanics and impact on stresses in the hub.
- Press w/ Torque: Press now transferring max driveline torque.
- Shock Torque: A sudden torque applied to the driveline. Happens when the driveline is spun up and the car is dropped on a high traction surface.
- Curb Slam X: The car taking a 700 lbs load into the tire sidewalls 11 from the center of the hub, while the hub is in an X shape.
- Curb Slam +: The car taking a 700 lbs load into the tire sidewall 11" from center while the hub is in a + shape. This load case has historically been the limiting load case.
- Press w/ Torque & Curb Slam +: New load Case that combines the old limiting + Curb Slam load case while the car is full throttle with the press accurately modeled.
- Breaking: Breaking forces applied to the front rotor towers.

1.5.2 Press w/ Torque & Curb Slam +

Worst possible loading condition for a hub is shock torque while undergoing a curb slam. This load case is chosen because it's not unreasonable for a hub to be shock torqued while undergoing a curb slam event. Imagine it as the car landing on one tire tilted while giving it full throttle.

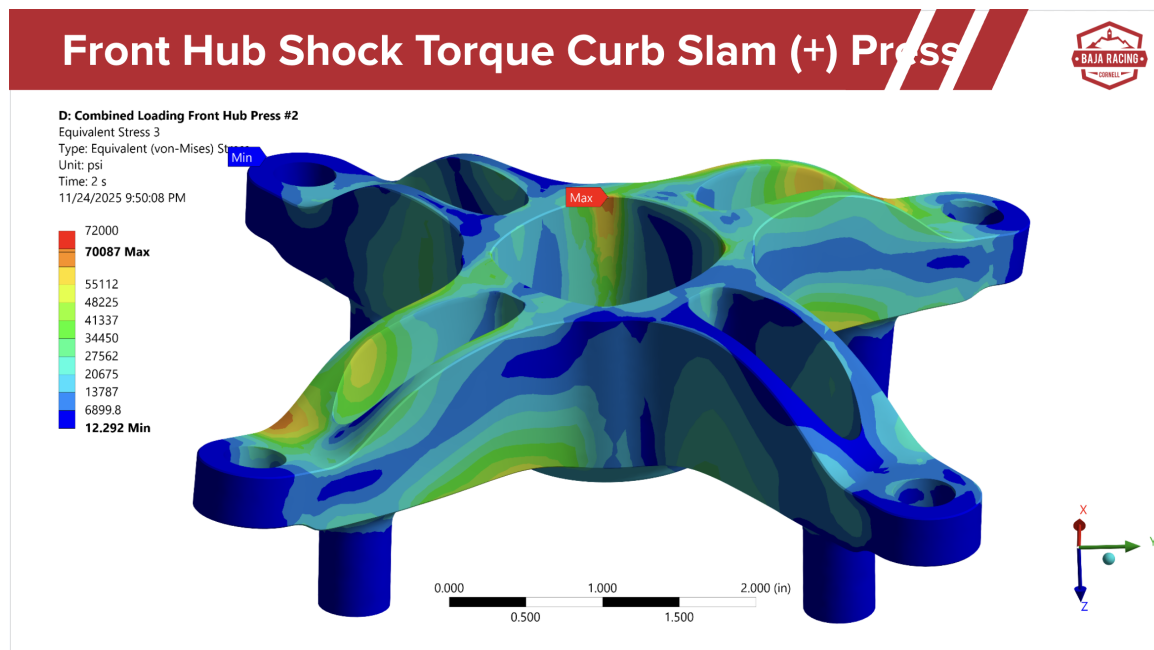


Figure 8: Front Worst Case loading condition

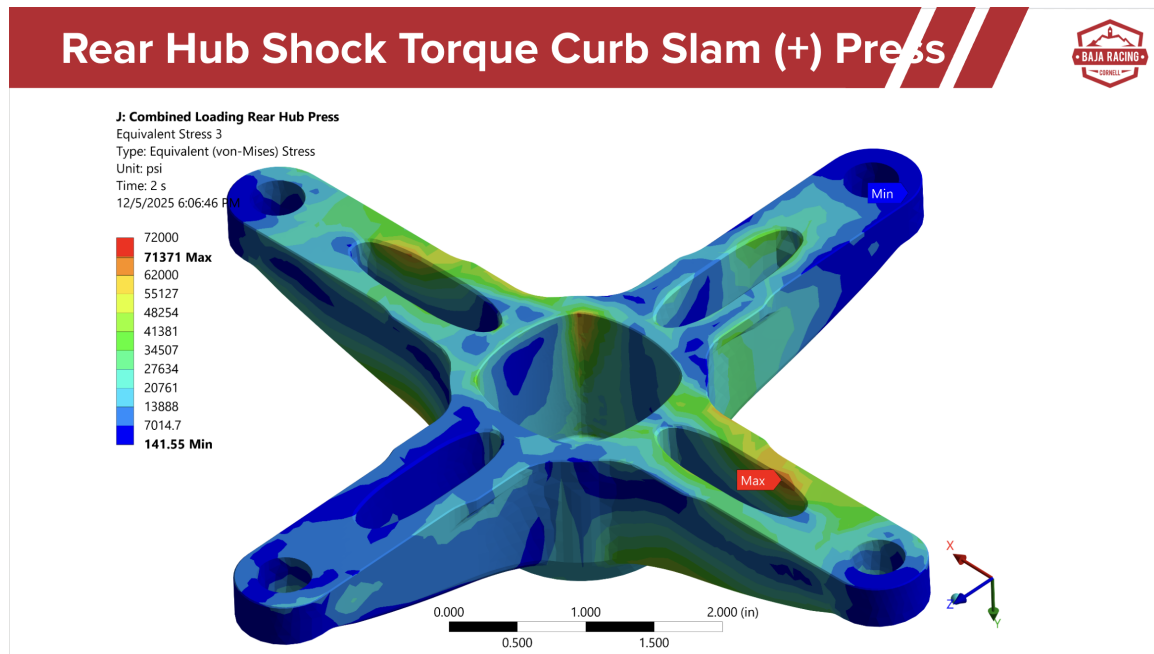


Figure 9: Rear Worst Case loading condition

2 Personal Reflection

I didn't realize until writing this report on how much I did this year and how much I learned. I really enjoyed the design work that I did and the work I accomplished this year feels really meaningful. I also had a blast machining stuff in the shop. Even though I did not go to Jan-Fab this year I was able to contribute later in the semester.